

Perineuromal hydrodissection for acute postamputation pain? An observational study in a time of war

WHAT IS ALREADY KNOWN ON THIS TOPIC There are no reliable methods to prevent or treat postamputation pain. Peripheral mechanisms, including neuromas, are thought to play a role for both residual limb and phantom limb pain.

WHAT THIS STUDY ADDS This exploratory study suggests that hydrodissection may be a viable approach to treat acute postamputation pain, with a potentially larger effect on residual limb than phantom limb pain. It also provides initial estimates of the treatment effect size to inform future research.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE, OR POLICY Although the effectiveness of hydrodissection was limited to certain populations, the safety, low cost, and ease of use warrant further investigation. In particular, this study suggests that future studies should focus on the treatment of acute pain and consider factors such as the duration of pain, as well as psychological and medication-related issues, when studying interventions for postamputation pain.

Introduction Amputations are a signature injury of war. In a recent systematic review that included 31 studies (n=14 738), the authors estimated the pooled prevalence rates of phantom limb pain (PLP) and residual limb pain (RLP) to be 57% and 61%, respectively, among combat-injured service members. 1 In addition to pain, amputation is associated with an increase in mortality and reduction in quality of life. 2 4 There are no reliable treatments for postamputation pain (PAP), which stems from the fact that it represents a syndrome rather than disease, with a myriad of different mechanisms. There is significant overlap in PLP and RLP, indicating the likelihood of overlapping mechanisms and risk factors. 5 These mechanisms include peripheral (eg, neuroma), spinal, and central etiologies (eg, central sensitization), with a presumed higher contribution of peripheral mechanisms influencing RLP. 3 For PLP, a review found level 2 evidence for opioids, gabapentin, and direct current stimulation and level 3 evidence for anti-depressants, N-methyl-D-aspartate receptor antagonists and peripheral nerve stimulation. 3 There are no systematic reviews for RLP, but small randomized trials support opioids, intravenous lidocaine, and nerve stimulation, with anecdotal evidence in favor of targeted muscle re-innervation, perineural injections, and ablation. 6 For persistent postsurgical pain in general, there is a correlation between severe acute postsurgical pain and chronification, with efforts to reduce chronic pain by mitigating acute pain yielding mixed results. 7 One treatment that has not been previously studied for PAP is hydrodissection. Hydrodissection involves the injection of fluid into the tissue planes surrounding nerves. In addition to releasing entrapped nerves, hydrodissection may also suppress ectopic nerve discharges, exert local anti-inflammatory effects by decreasing cytokine levels, prevent or even wash out early heterotopic ossifications, block nociceptive signaling, and promote tissue healing. When dextrose solutions are added, it may reverse the hypoglycemic milieu around ischemic nerves that sensitize c-fibers. 8 11 Randomized and uncontrolled trials examining hydrodissection support its efficacy for entrapment syndromes and cervical radiculopathy, with emerging evidence for other conditions. 11 13 In the Ukraine-Russian war, it is estimated that over 100 000 Ukrainians have sustained war-related amputations. Unfortunately, there remains an absence of data on disease burden, treatments or outcomes. 14 Prewar, Ukraine's economy ranked 59th in the world, with less than 8% devoted to healthcare translating to per capita spending of less than \$300 per person. This renders advanced treatments inaccessible to most war casualties, while therapies such as opioids carry risks in a population with high co-prevalence rates of psychopathology. 15 To evaluate the effect of hydrodissection on acute PAP, we performed an observational study comparing hydrodissection and opioids to opioids alone. Our objectives were: (1) to determine the effectiveness of hydrodissection as an add-on therapy to opioids for phantom limb and residual limb pain; (2) determine whether hydrodissection reduces the incidence of persistent PAP; and (3) identify factors associated with treatment response. Patients Permission to conduct this observational study was granted by the Institutional Review Board at Vinnytsia National Pirogov Medical University (IRB_LEC_5\13052022) and the Public Affairs Department at Walter Reed National Military Medical Center. On 13 May 2022, it was registered as a database designed to collect data on war-related chronic pain conditions with the central government committee (<https://nddkr.ukrintei.ua/>, 0122U002551). All patients received treatments that were already being provided as standard clinical care and provided informed written consent. Seventy-four patients with PAP treated with either oral opioids alone or opioids and ultrasound-guided hydrodissection between 1 November 2023 and 1 April 2024 at Vinnitsa Regional Clinical Hospital or Prometheus Pain Treatment and Rehabilitation Center were evaluated. Analysis criteria were major limb amputation; treatment with opioids with or without hydrodissection; stable dose of opioid and other analgesics for 1 week; numerical rating scale pain score 4/10 for either RLP or PLP over the past week; duration of pain between 2 weeks and 6 months; and failure to respond to physical therapy. Excluded from analyses were those with systemic or local residual limb infection or an open wound that precluded a block and represented a treatable pain generator; patients missing 4-week follow-up data; previous procedure within 4 weeks of hydrodissection; pain in other location greater than RLP and PLP; and concomitant injection therapy received before 4-week follow-up. Nerve hydrodissection

technique Painful neuromas were identified by history (eg, determining whether phantom pain occurred in a specific nerve distribution) and physical exam (eg, Tinel's sign). Depending on the clinical presentation, either one or two end neuromas were identified using ultrasound guidance. These included the sciatic and femoral nerves proper and their main branches, the common peroneal, tibial, and saphenous nerves. After anesthetizing the skin with lidocaine 1%, a 21- or 22-gauge ultrasound needle was inserted in-plane. In cases where the end neuroma was obscured by scar tissue, an out-of-plane technique was employed. Once the needle tip appeared near the neuroma, a 5% dextrose in water solution was injected, with total volumes ranging between 5 and 40 mL depending on the number and diameter of targeted nerves and the extent of encasing scar tissue. Initial opioid titration with immediate-release morphine or oxycodone was performed by an evaluating physician starting at 10 mg oral morphine equivalents every 612 hours as needed and titrating up or down at 3- to 7-day intervals based on patient response. There were no limits on opioid dose, and opioid stability was defined as no changes in opioid dose or of another analgesic for at least 1 week. No patient was started on adjuvants during the study period, but doses of previously prescribed adjuvants were titrated as clinically indicated. Outcome measures Baseline demographic and clinical characteristics including location of amputation(s), average and worst PLP and RLP scores, Hospital Anxiety and Depression Scale (HADS) score, Somatic Symptom Scale-8 (SSS-8) V.2 score, analgesic medications, number of nerves and limbs targeted, and volume of injectate were recorded. For opioid dose, oral morphine equivalents per day were recorded based on equianalgesic conversion. In between the procedure and follow-ups, no additional procedures were performed, though patients could receive the non-steroidal anti-inflammatory drug dexamethasone as needed, titration of pre-treatment adjuvants, or additional physical therapy. Follow-up evaluations occurred at the 48-hour, 1-week, 4-week, and 12-week time points. The coprimary outcome measures were median reduction from baseline in average RLP and PLP scores at the 12-week endpoint. Secondary outcome measures included change and absolute scores for RLP and PLP at earlier time points; change and absolute scores for HADS and SSS-8 at follow-up; medication reduction (eg, opioid dose and a categorical medication reduction outcome defined as 20% reduction in opioid consumption) at follow-up 16 ; satisfaction with the procedure rated as highly satisfied, somewhat satisfied, or dissatisfied; and a binary outcome measure whereby success was defined as a 2-point or 30% improvement in average RLP and PLP score, being at least somewhat satisfied with the procedures results, and not requiring an increase in analgesic consumption (20% increase in opioid dose or starting a new non-opioid analgesic). Statistical analysis Our sample size had sufficient power to detect a minimal clinically important difference of 1.0 point between the opioid-only and hydrodissection + opioid treatment groups based on the following assumptions: baseline average pain score of 5.0; a medium-to-large effect size of 0.7; () level of 0.05; and statistical power 80%. 17 20 As no preliminary data were available on the use of hydrodissection + opioids for PAP, the effect size was estimated based on prior studies of hydrodissection suggesting a large effect. Based on these assumptions, we estimated that 37 patients in each treatment group were needed. After dropouts, we achieved final sample sizes of 36 patients in the opioid-only group and 38 patients in the hydrodissection and opioid group, providing 85% power. In subgroup analyses, patients were stratified by duration of pain into acute (1 month, 2 months, and 3 months) or chronic (3 months). All analyses were performed in RStudio 2024.04.01. 2 Fische's exact (if n 5) tests were applied for comparisons between hydrodissection+opioid versus opioid-only categorical outcomes. For continuous variables, Wilcoxon rank-sum was used. Frequency (%) and mean (SD) were reported for variables that met normality, while median (IQR) was reported for variables that did not per Shapiro-Wilk tests, Q-Q plots, and frequency histograms. Results Between 1 November 2023 and 1 April 2024, 92 patients with extremity amputations were screened. Fifteen patients were excluded for infection over the site (n=4), open wound (n=2), cognitive impairment (n=8), or refusal of participation (n=1), with another three excluded for failure to report 4-week outcomes. In sum, 38 patients received both hydrodissection+opioids, and 36 patients received opioids only (figure 1). These patients all completed 12-week follow-up. Figure 1 Study flow chart. Baseline characteristics differed for some variables between the two treatment groups (table 1). Patients receiving hydrodissection+opioids were younger than the opioid-only group (mean±SD age: 35.13±9.50 vs . 40.39±9.38 years old, p=0.02), had a shorter duration of pain (median (IQR): 2.00±1.88 vs . 3.75±3.00 months, p 0.01), and used less opioids at baseline (p 0.01). Patients receiving only opioids were more likely to be receiving antidepressants (86% vs 34%), gabapentin (53% vs 0%), and non-steroidal anti-inflammatory drugs (100% vs 89%) (p 0.001). Injuries in the hydrodissection + opioid group were more likely to result in unilateral amputations above the knee (63% vs 36%, p 0.01). There was no difference between treatment allocation and outcomes stratified by military status (data redacted). Table 1 Baseline demographic and clinical characteristics of study subjects

Characteristic

	* Hydrodissection+opioids (n=38)	Opioids only (n=36)	Total cohort (n=74)
Age, years (mean±SD)	35.13±9.50	40.39±9.38	37.69±9.75
Male (n, %)	38/38 (100)	36/36 (100)	74/74 (100)
Duration of pain (months, median (IQR))	2.00 (1.88)	3.75 (3.00)	2.00 (2.50)
Median opioid dose (morphine milliequivalents/d, mean±SD)	41.32±9.63	58.19±28.64	49.53(22.64)
Smoking (n, %)	37/38 (97)	35/36 (97)	72/74 (97)
Etiology (n, %)	Blast injury 30/38 (79)	35/36 (97)	65/74

(88) Missile injury 8/38 (21) 1/36 (3) 9/74 (12) Amputation location (n, %) Unilateral above knee 13/38 (34) 10/36 (28) 23/74 (31) Unilateral below knee 24/38 (63) 13/36 (36) 37/74 (50) Bilateral above knee 1/38 (3) 5/36 (14) 6/74 (8) Upper extremity 0/38 (0) 8/36 (22) 8/74 (11) Co-existing psychiatric condition (n, %) Depression 12/38 (32) 7/36 (19) 19/74 (26) Anxiety 35/38 (92) 36/36 (100) 71/74 (96) Posttraumatic stress disorder 33/38 (87) 32/36 (88) 65/74 (88) Traumatic brain injury 0/38 (0) 0/36 (0) 0/74 (0) Multiple psychiatric disorders 35/38 (92) 32/36 (88) 67/74 (91) None 0/38 (0) 0/36 (0) 0/74 (0) Adjuvants Antidepressants 13/38 (34) 31/36 (86) 44/74 (59) Gabapentin 0/38 (0) 19/36 (53) 19/74 (26) Non-steroidal anti-inflammatory drugs 34/38 (89) 36/36 (100) 70/74 (95) Average residual limb pain score (010 NRS, median (IQR)) 6.00 (2.00) 6.00 (1.00) 6.00 (1.00) Worst residual limb pain score (010 NRS, median (IQR)) 8.00 (1.00) 8.00 (1.00) 8.00 (1.00) Average phantom limb pain score (010 NRS, median (IQR)) 5.00 (1.00) 5.00 (1.00) 5.00 (1.00) Worst phantom limb pain score (010 NRS, median (IQR)) 5.00 (1.00) 6.00 (1.00) 5.50 (1.00) Hospital Anxiety Scale score (median (IQR)) 10.50 (2.75) 10.00 (2.00) 10.00 (2.00) Hospital Depression Scale score (median (IQR)) 8.00 (4.00) 8.00 (2.00) 8.00 (3.00) Somatic Symptom Scale-8 score (median (IQR)) 13.50 (6.75) 13.00 (5.25) 13.00 (6.00) Hydrodissection volume (total mL, mean±SD) 19.29±3.41 * All patients also received non-steroidal anti-inflammatory drugs and/or adjuvants. NRS, numerical rating scale. Primary outcome measures Primary outcome measures are listed in table 2 . At 12 weeks, the hydrodissection + opioid group had lower average RLP scores (3.50±2.00 vs 5.00±0.00, p 0.001), greater reduction in average RLP scores from baseline (2.00±1.00 vs 1.00±1.00, p 0.001), and lower worst RLP scores (4.00±2.00 vs 5.00±0.25, p 0.01). There was no significant difference between the two treatment groups for the median change in average or worst PLP scores. Table 2 Continuous outcome measures stratified by treatment group

Outcome measure	Hydrodissection+opioids (n=38)	Opioids only (n=36)	P-value *
48-hour average RLP (010 NRS score, median (IQR))	4.00 (2.75)	6.00 (1.25)	0.001
48-hour worst RLP (010 NRS score, median (IQR))	5.00 (2.00)	6.00 (2.00)	0.01
48-hour average PLP (010 NRS score, median (IQR))	5.00 (1.00)	5.00 (1.00)	0.02
48-hour worst PLP (010 NRS score, median (IQR))	5.00 (1.00)	5.00 (0.00)	0.01
48-hour Hospital Anxiety Scale score (median (IQR))	9.00 (2.75)	9.50 (1.00)	0.33
48-hour Hospital Depression Scale score (median (IQR))	8.00 (1.00)	8.00 (1.25)	0.67
48-hour Somatic Symptom Scale-8 score (median (IQR))	14.00 (5.75)	13.00 (3.75)	0.20
1-week average RLP score (median (IQR))	3.00 (2.00)	5.00 (1.00)	0.001
1-week worst RLP score (median (IQR))	5.00 (1.00)	5.00 (0.00)	0.01
1-week average PLP score (median (IQR))	4.00 (0.75)	4.00 (1.00)	0.03
1-week worst PLP score (median (IQR))	5.00 (1.00)	4.00 (1.00)	0.78
1-week Hospital Anxiety Scale score (median (IQR))	9.00 (2.75)	10.00 (2.00)	0.01
1-week Hospital Depression Scale score (median (IQR))	8.00 (0.00)	8.00 (1.00)	0.10
1-week Somatic Symptom Scale-8 score (median (IQR))	12.00 (2.75)	13.00 (2.00)	0.45
4-week average RLP score (median (IQR))	3.00 (3.00)	5.00 (1.00)	0.001
Change in 4-week average RLP score from baseline (median (IQR))	2.00 (2.00)	0.50 (1.00)	0.001
4-week worst RLP score (median (IQR))	4.50 (1.00)	5.00 (0.00)	0.001
Change in 4-week worst RLP score from baseline (median (IQR))	3.00 (1.00)	3.00 (1.00)	0.01
4-week average PLP score (010 NRS score, median (IQR))	4.00 (1.00)	5.00 (1.00)	0.001
Change in 4-week average PLP score from baseline (median (IQR))	0.00 (1.00)	+1.00 (1.00)	0.001
4-week worst PLP score (010 NRS score, median (IQR))	5.00 (1.00)	5.00 (1.00)	0.44
Change in 4-week worst PLP score from baseline (median (IQR))	-1.00 (1.00)	-1.00 (2.00)	0.78
4-week Hospital Anxiety Scale score (median (IQR))	10.00 (3.00)	11.00 (1.00)	0.001
4-week Hospital Depression Scale score (median (IQR))	8.00 (1.00)	9.50 (2.00)	0.01
4-week Somatic Symptom Scale-8 score (median (IQR))	12.00 (2.00)	13.00 (2.00)	0.17
12-week average RLP score (median (IQR))	3.50 (2.00)	5.00 (0.00)	0.001
Change in 12-week average RLP score from baseline	2.00 (1.00)	1.00 (1.00)	0.001
12-week worst RLP score (median (IQR))	4.00 (2.00)	5.00 (0.25)	0.001
Change in 12-week worst RLP score from baseline (median (IQR))	3.50 (2.00)	3.00 (1.00)	0.008
12-week average PLP score (median (IQR))	5.00 (1.00)	5.00 (1.00)	0.01
Change in 12-week average PLP score from baseline (median (IQR))	0.00 (2.00)	1.00 (1.00)	0.17
12-week worst PLP score (median (IQR))	5.00 (1.00)	5.00 (1.00)	0.92
Change in 12-week worst PLP score from baseline (median (IQR))	-1.00 (1.00)	-1.00 (1.25)	0.35
12-week Hospital Anxiety Scale score (median (IQR))	10.00 (2.00)	11.00 (1.00)	0.001
12-week Hospital Depression Scale score (median (IQR))	9.50 (1.00)	9.00 (1.00)	0.41
12-week Somatic Symptom Scale-8 score (median (IQR))	12.00 (2.00)	13.00 (1.50)	0.03

* P values were calculated using the Wilcoxon rank-sum test (Mann-Whitney U test) comparing continuous outcomes between hydrodissection and opioids versus opioid-only groups. NRS, numerical rating scale; PLP, phantom limb pain; RLP, residual limb pain. Secondary outcome measures At 48 hours post-treatment, the hydrodissection+opioid group had lower average RLP (4.00±2.75 vs 6.00±1.25, p 0.001) and worst RLP scores (5.00±2.00 vs 6.00±2.00, p 0.01). One-week post-treatment, the hydrodissection group had lower average RLP (3.00±2.00 vs 5.00 1.00, p 0.001) and worst RLP scores (5.00±1.00 vs 5.00±0.00, p 0.01) as well. At 4 weeks, average RLP was lower (3.00±3.00 vs 5.00±1.00, p 0.001), and reduction in average RLP score was greater (2.00±2.00 vs 0.50±1.00, p 0.001) in the hydrodissection group. For RLP, the proportion of patients with a categorical positive outcome was higher in the hydrodissection group than the opioid-only group across all timepoints. There was no significant difference between groups at any timepoint for PLP (figure 2). Figure 2 Percent positive categorical outcome for

residual limb or phantom limb pain stratified by treatment group. Positive outcome designated as 2-point or 30% decrease in average residual limb or phantom limb, being at least somewhat satisfied with treatment, and not requiring an increase in analgesic usage. Vertical lines represent the 95% CI. Asterisks (*) denote significant differences between groups (p 0.05). Aside from 12 weeks where the hydrodissection+opioid group had lower SSS-8 scores compared with the opioid-only group (12.00 (2.00) vs 13.50 (1.50), p=0.03), there were no significant differences in SSS-8 scores at other timepoints. HADS depression domain score was not significantly different between treatment groups at any timepoint except 4 weeks when patients receiving hydrodissection had lower depression subscale scores (8.00±1.00 vs 9.50±2.00, p 0.01). Patients in the hydrodissection group had lower HADS anxiety subscale scores at 4 weeks (10.00±3.00 vs 11.00±1.00, p 0.001) and 12 weeks (10.00±2.00 vs 11.00±1.00, p 0.001). Patients who received opioids only started at a higher morphine equivalent dose than those who received hydrodissection (58.19±28.64 vs 41.32±9.63; p 0.01). The hydrodissection group reduced their opioid consumption at 12-week follow-up compared with baseline (41.32±9.63 to 33.42±17.50 morphine equivalents per day; p=0.001), while the opioid-only group did not. At 48 hours, 1 week, and 4 weeks, there were a higher proportion of patients in the hydrodissection group that had positive opioid reduction (figure 3). Figure 3 (A) Box and whisker plots of opioid dose (morphine equivalents per day) at each study timepoint, stratified by treatment group. Boxes represent the IQR, with the median indicated by the horizontal line. Whiskers extend to the most extreme data points within 1.5 times the IQR from the 25th percentile (Q1) and 75th percentile (Q3). Outliers are shown as individual points beyond the whiskers. Asterisks (*) denote significant differences between groups (p-value 0.05). (B) Percent positive analgesic medication reduction outcome at each study timepoint, stratified by treatment group. Positive opioid dose reduction outcome defined as cessation of non-opioid analgesic or 20% reduction in opioid consumption compared with baseline. Asterisks (*) denote significant differences between groups (p-value 0.05). For satisfaction, 24% and 32% of patients who received opioids and hydrodissection were highly and somewhat satisfied with the treatment results at 12 weeks, respectively, versus 14% and 44% in the opioid-only group (p=0.44 for highly satisfied, p=0.37 for somewhat satisfied). No patient in either group reported a complete absence of RLP or PLP at 12 weeks. Neither did any patient report a 12-week average RLP score 2, though one patient, in the opioid-only group, reported a 12-week average PLP score of 2/10. Effect of pain acuity on main outcome measures When patients were stratified into acute (3-month duration) and chronic (3 months) pain, differences emerged. The hydrodissection+opioid group patients with acute pain had lower 4-week (median 3.0 IQR 3.0 vs 5.0 (2.0); p=0.001) and 12-week (3.5 (2.0) vs 5.0 (0.0); p=0.001) average RLP pain scores than the opioid-only group, but the difference between groups among those with chronic pain did not reach statistical significance. For average PLP, the difference among those with acute pain was significant at 4 weeks (4.0 (1.0) vs 5.0 (2.0); p=0.04) but fell shy of significance at 12 weeks. In the chronic pain subgroup, this same pattern of significance at 4 weeks (4.5 (2.0) vs 5.0 (0.0); p=0.01) but not at 12 weeks held true (table 3).

Variable	Hydrodissection+opioids (n=38)	Opioids only (n=36)	P-values
Average RLP (010 NRS score, median (IQR))			
Baseline	6.0 (2.0)	5.5 (1.25)	0.31
4 weeks	3.0 (3.0)	4.0 (2.50)	0.01
12 weeks	3.5 (2.0)	4.0 (2.25)	0.07
Positive 12-week RLP categorical outcome (n/N, %)	15/30 (50%)	4/8 (50%)	0.01
Average PLP (010 NRS score, median (IQR))			
Baseline	5.0 (1.0)	5.0 (1.5)	0.04
4 weeks	4.0 (1.0)	4.5 (2.0)	0.01
12 weeks	4.0 (1.0)	5.0 (0.0)	0.13
Positive 12-week PLP categorical outcome (n/N, %)	1/30 (3.3%)	0/8 (0%)	0.01
Worst RLP (010 NRS score, median (IQR))			
Baseline	8.0 (1.0)	8.0 (1.25)	0.83
4 weeks	5.0 (1.0)	5.0 (0.0)	0.001
12 weeks	4.0 (2.0)	4.0 (0.5)	0.001
Worst PLP (010 NRS score, median (IQR))			
Baseline	5.0 (1.0)	6.0 (1.0)	0.59
4 weeks	5.0 (1.0)	5.0 (1.0)	0.74
12 weeks	5.0 (0.75)	5.0 (1.0)	1.00
HADS combined score (042, median (IQR))			
Baseline	18 (2.75)	21 (5.5)	0.13
4 weeks	19 (3.00)	21 (2.75)	0.02
12 weeks	19 (3.00)	21 (2.75)	0.02
Positive 12-week HADS categorical outcome (n/N, %)	0/30 (0%)	0/8 (0%)	1.00
SSS-8 score (032, median (IQR))			
Baseline	13 (3.75)	18 (7.5)	0.61
4 weeks	12 (2.0)	11.5 (4.0)	0.24
12 weeks	12 (2.0)	11.5 (4.0)	0.24
Positive 12-week SSS-8 categorical outcome (n/N, %)	§ 10/30 (33%)	6/8 (75%)	0.04
Opioid dose (oral morphine equivalents per day, mean±SD)			
Baseline	40.33±9.64	45.00±9.26	0.01
4 weeks	23.00±15.35	25.00±16.90	0.01
12 weeks	33.33±8.84	33.75±9.16	0.09
Positive 12-week opioid reduction categorical outcome (N, %)	¶ 12/30 (40)	5/8 (63)	0.01

* Top line p-value represents statistical difference between hydrodissection+opioids versus opioids-only for a 3-month duration (ie, acute pain definition), while second line represents difference for a 3-month duration (ie, chronic pain). For continuous variables, comparisons between the hydrodissection+opioids group and the opioid-only group for pain duration stratification were performed using the

Wilcoxon rank-sum test. For categorical variables, 2 tests were used, or Fishers exact test if any expected cell count was 5. Positive categorical outcome designated as a 2-point or 30% reduction in pain from baseline at 12 weeks, being at least somewhat satisfied with treatment, and not requiring an increase in analgesics (20% opioid increase or starting a non-opioid). Positive categorical outcome designated as a 6-point reduction from baseline. § Positive categorical outcome designated as a 3-point reduction from baseline. ¶ Positive categorical outcome designated as a 20% reduction in opioid dose from baseline. HADS, Hospital Anxiety and Depression Scale; NRS, numerical rating scale; PLP, phantom limb pain; RLP, residual limb pain; SSS-8, Somatic Symptom Scale-8. Patients with acute pain were further stratified by duration of pain into 24-week, 48-week, and 812-week groups. In patients who were 812 weeks out from pain onset, the hydrodissection + opioid group had a greater difference in average RLP from baseline at 12 weeks (2.0 (1.0) vs 0.5 (1.5), p 0.04). None of the outcomes were significant for PLP. Among patients who were 24 weeks out since pain onset, those in the hydrodissection + opioid group had lower worst RLP scores at 4-week study follow-up than those in the opioid-only group (4.0 (1.0) vs 5.0 (0.5), $p=0.048$). All other outcomes were not statistically significant. Adverse events In the hydrodissection group, one patient developed what appeared to be new-onset phantom pain in his amputated leg, which resolved over several weeks. Three patients in the opioid-only group developed a diffuse pain phenotype and sensitivity over the stump, indicative of opioid-induced hyperalgesia versus possible complex regional pain syndrome. Discussion The main finding of this study was that hydrodissection was associated with a significant reduction in RLP and less influence on PLP across the study, though unequal baseline measures including age, duration of pain, and opioid dose limit the conclusions one can draw on effectiveness. For average RLP, this effect was only significant in amputees with acute pain. Comparison to other studies There are no studies evaluating hydrodissection for PAP, but clinical studies have reported effectiveness for neuropathic pain, though the addition of corticosteroid to a high-volume injectate in one double-blind study may have contributed to benefit. 12 17 18 For acute pain, one randomized trial that added hydrodissection to local anesthetic and distilled water during an infraclavicular block reported less postoperative pain in the hydrodissection group. 21 High postoperative pain is a strong predictor for persistent postsurgical pain, including after amputation. 7 22 23 Explanation of findings Our results are cautiously optimistic, with hydrodissection+opioids reducing RLP and to a lesser degree PLP scores compared with opioids only. For the coprimary outcome measures (average RLP and PLP at 12 weeks) and categorical outcomes stratified by acuity, patients in the hydrodissection group with a duration of pain 3 months experienced greater decreases in average RLP score than those in the opioid-only group and those with chronic pain. Disease burden in the opioid-only group was greater on many counts than that observed in the hydrodissection group. In the pain literature, measures of disease burden such as duration of pain, opioid consumption, and pain scores are associated with negative outcomes. 7 24 In patients whose PAP was due to lingering postoperative pain, which likely occurred more often in the hydrodissection group, contributions from the natural course of healing may partially explain stronger responses. Putative hydrodissection mechanisms As noted above, there are numerous mechanisms by which hydrodissection may alleviate pain. 8 10 Since PAP is a syndrome with disparate mechanisms, it is impossible to tease out which mechanisms, if any, were addressed by hydrodissection. However, the observation that positive outcomes were more frequent in RLP than for PLP, and that cortical mechanisms are more relevant for PLP than RLP, suggests peripheral mechanisms. 3 It is also possible that the peripheral and systemic treatment of PAP in the forms of hydrodissection and opioid therapy, respectively, beneficially alter cortical reorganization, a likely mechanism for PLP. 25 26 The contribution of a placebo effect cannot be discounted. The placebo effect is stronger for pain and psychiatric indications (which are highly coprevalent) than for more objective outcome measures, being at least as great as the intrinsic effect for most therapies. 27 Moreover, this effect is larger for procedures such as hydrodissection than for medications. 28 It is therefore probable that even an interventional treatment that lacked efficacy would be more effective than having no procedure in an uncontrolled study. In one systematic review evaluating long-standing perineural local anesthetic infusions for PAP, Laloo et al found a modest effect on postoperative pain and opioid requirements when all studies were evaluated, but the effect on opioid requirements was no longer significant when only randomized trials were included. 29 Confounding effects of psychiatric variables, amputation burden, pain duration, opioid dose, and adjuvant use As alluded to above, there were baseline differences in duration, opioid dose, adjuvant usage, amputation classification, and psychiatric variables that may have played a role in outcomes. In clinical studies, longer duration of pain is consistently associated with poor treatment results. 7 24 Although one systematic review on PAP found only small differences in prevalence rates of RLP and PLP stratified by duration, the authors did not sub-analyze results for shorter durations of pain, and chronic pain tends to be stable after 6 months. 1 An observational study comparing PLP prevalence in 85 amputees found the highest incidence at the earliest surveillance point at 6 months, declining slightly over the 3-year follow-up. 30 A cross-sectional survey study found that 48% of 65 amputees reported that both RLP and PLP decreased over time. 31 In our study, the lack of a significant effect for RLP in the chronic pain cohort may signify that the postinjury inflammatory

phase returned to baseline. Other explanations include more intractable scar tissue less amenable to hydrodissection, smaller numbers, lower power to detect a difference, and collinearity with other variables accounted and unaccounted (greater attention and number of visits for newer patients) for. There were higher baseline opioid doses in the opioid-only group. This may be because those with a longer duration of pain were more refractory to treatment or had more time to develop tolerance. Opioids have been shown to strongly predict the chronification of pain after iatrogenic and non-iatrogenic traumatic injuries, including during war, as well as procedural treatment failure. 7 32 33 Reasons might include opioid hyperalgesia, conscious or subconscious secondary gain, non-pain-related use, and that higher opioid doses tend to be prescribed for those who have failed multiple other therapies, especially in veterans who engender a more tolerant approach than what guidelines recommend. 7 34 Explanations as to why opioid use did not predict treatment results are that they also effectively reduced pain, anxiety, and sleep disturbances, the latter of which are associated with poor outcomes, and that not enough time elapsed before hyperalgesia and dependence developed. 7 24 There were a higher proportion of hydrodissection patients who had unilateral lower extremity amputations compared with those in the opioid-only group. This might have accounted for part of the reason why hydrodissection was not used in the opioid-only group (ie, procedures are best utilized in those with more localized pain), but also could pose as a confounding variable to explain the poorer outcomes in the opioid-only group (ie, greater disease burden translates to poorer outcomes). 35 36 Adjuvant use, particularly gabapentin usage which was 0% in the hydrodissection+opioid treatment group, was higher in the group that did not undergo hydrodissection. This may reflect the desire for non-hydrodissection patients to avoid procedures, or the reluctance of practitioners to trial multiple interventions simultaneously in an attempt to determine individual effectiveness. Some studies have found adjuvant use to predict poor interventional treatment outcomes, which may reflect therapeutic recalcitrance. 37 Other studies suggest that gabapentin may work better for neuropathic pain phenotypes associated with central sensitization (dynamic mechanical hyperalgesia), which we did not measure, but is more likely to be associated with long-standing pain. 38 Finally, we did not control for psychiatric comorbidities which are highly associated with pain outcomes. 7 24 Limitations and future research There are limitations to this study that warrant consideration. First, the design resulted in differences in baseline variables that could adversely affect outcome including the possibility of unmeasured variables such as catastrophization, coping skills, central sensitization, and opioid use disorder. Those in the hydrodissection group also were younger, which is independently associated with persistent postsurgical pain. 7 Second, the placebo effect may have played an outsized role in the hydrodissection group, who probably had higher expectations. This is supported by the observation that pain itself decreased more in the treatment group than secondary outcome measures such as anxiety that tend to move in concert with pain. Recording objective outcome measures such as pain pressure thresholds, activity levels, or biomarkers in future studies may provide better information on outcomes and mechanisms. Third, hydrodissection is likely to work better for pain secondary to neuroma formation rather than central mechanisms. Whereas both RLP and PLP can be secondary to neuromas, treatments that address peripheral pathology, including inflammation and ectopic discharges, may be more likely to treat the former. 39 Future studies should consider identifying painful neuromas via diagnostic blocks or adding local anesthetic and/or steroid to the injectate to reduce spontaneous neuroma firing and decrease the inflammatory milieu in early amputation. Finally, the treatments compared were not standardized, and some non-procedural co-interventions such as non-steroidal anti-inflammatory drugs and physical therapies were permitted which could have confounded results. Conclusions In this study, hydrodissection added to opioids improved PAP, particularly RLP, compared with opioids only. However, this effect was primarily driven by better outcomes in those with acute pain who were disproportionately represented in the hydrodissection group. Although these results suggest hydrodissection might benefit a subpopulation with PAP, randomized trials that control for confounding variables are needed to demonstrate efficacy.